



Abstract Class in Abstraction:

1. Abstract class it means partial class.
2. If class have any one method is abstract then class must have abstract.
3. Abstract class have both type of method (Implemented "Concrete" and Unimplemented "Abstract" methods).
4. If class have all implemented method only then also, we can make a class as an abstract.
5. We cannot create the object of abstract class but we can able to create the object reference of an abstract class by the help of their extended class.

Example: - Let's take a car, bike or train these are the transportation ways to travel we have some properties in common so it's important to have a transportation abstract class so that we use some methods in this abstract class that can be implemented in the derived or sub-classes.



CJC

Complete Java Classes

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These allow flexibility and reusable code which we can implement in further classes.

Abstract class

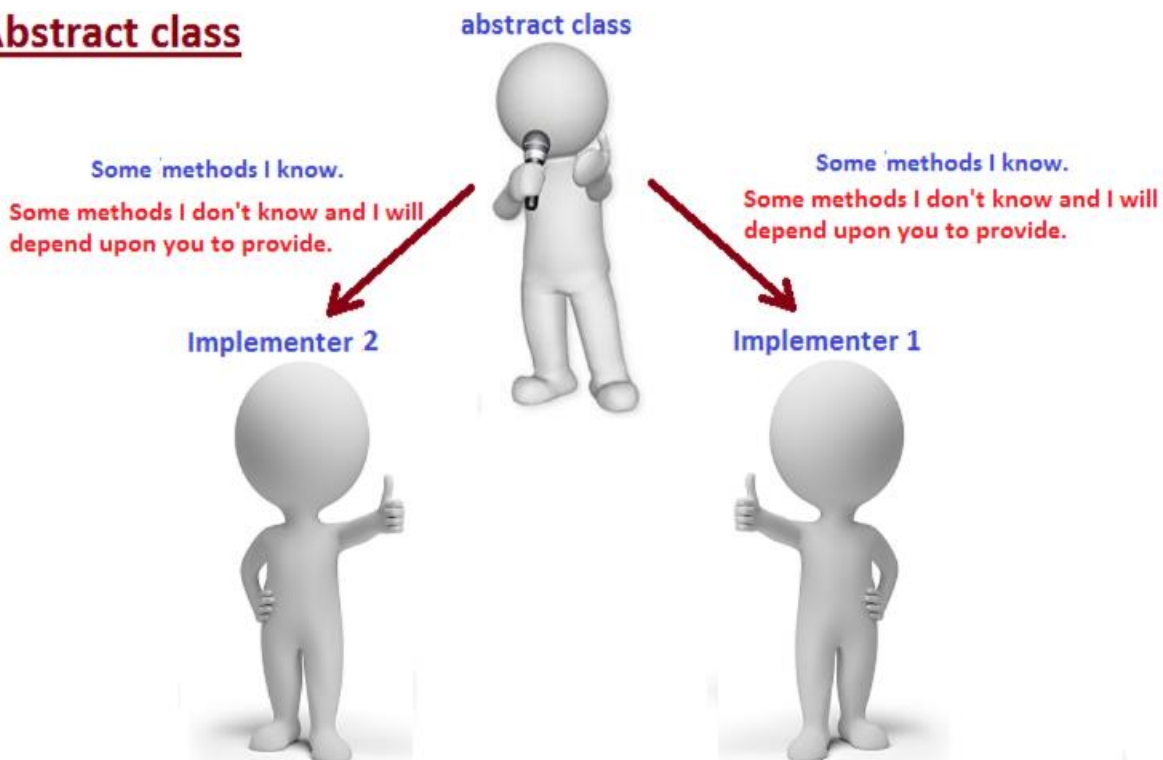


diagram of abstract class

Syntax of Abstract Class:

```
public abstract class ClassName {  
    //Implemented or Concrete Methods
```



//Un-implemented or Abstract Methods

}

What is the difference between abstract class and interface?

There are following difference them is given below-

Abstract Class	Interface
Declared using the keyword "abstract" .	Declared using the keyword "interface" .
Abstract class can be inherited using the keyword "extend" .	Interface can be inherited using the keyword "implements" .
Abstract class does not support multiple inheritance.	Interface supports multiple inheritance.
Abstract class can have both abstract and non-abstract methods.	Interface can only have abstract methods.
Supports final, non-final, static, non-static variables.	Supports only static and final variables.



What is the difference between abstract class and concrete class?

There are following difference them is given below-

Abstract Class	Concrete Class
An abstract class is declared using abstract keyword.	A concrete class is not declared using abstract keyword.
An abstract class cannot be directly instantiated using the new keyword.	A concrete class can be directly instantiated using the new keyword.
An abstract class may or may not contain abstract methods.	A concrete class cannot contain an abstract method.
An abstract class cannot be declared as final.	A concrete class can be declared as final.

Abstract Keyword: Abstract keyword in java is used to achieve abstraction. An abstraction is basically displaying only the details that are relevant to the end user. Using abstract keyword abstract classes and method can be created.